WORKED SOLUTION

STEP I 1993, Question 12

Mechanics • Projectile Interception

STEP I 1993, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q12

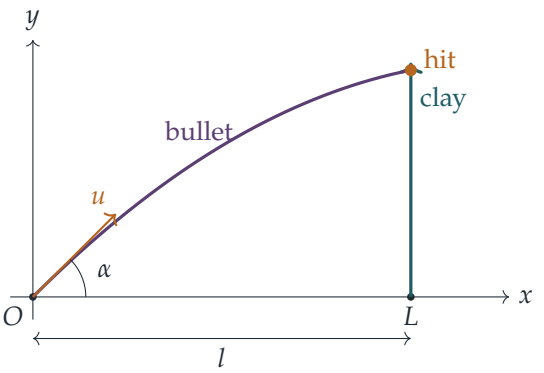
The key idea

A hit means the bullet and the clay are at the same place at the same moment. The bullet reaches the vertical line through the launcher at a time fixed by its horizontal motion alone, so we find that time first and then match the two heights. Gravity pulls both objects down by exactly the same amount, $\frac{1}{2}gt^2$, over the bullet's flight, so it cancels and leaves a tidy linear condition. When $T = 0$ this cancellation gives a neat way to read the whole problem.

Solution

The model

We treat the clay and the bullet as particles moving freely under gravity, with no air resistance, and assume the competitor fires in the vertical plane containing the launcher, towards it. Put the competitor at the origin O, with the x-axis horizontal towards the launcher L at $x = l$ and the y-axis vertically up. Measure time t from the moment the rifle is fired, so the clay has already been in the air for time T, and at time t it has been flying for $T + t$.



The condition for a hit

At time t after firing the bullet is at

$$x_B = (u\cos\alpha)\,t, \qquad y_B = (u\sin\alpha)\,t - \tfrac{1}{2}gt^2,$$

and the clay, which started up at speed v a time T earlier, is at $x = l$ with height

$$y_C = v(T+t) - \tfrac{1}{2}g(T+t)^2 = \left(vT - \tfrac{1}{2}gT^2\right) + (v - gT)\,t - \tfrac{1}{2}gt^2.$$

The bullet can only meet the clay on the line $x = l$, and it gets there when $(u\cos\alpha)\,t = l$, that is, at

$$t = \frac{l}{u\cos\alpha}.$$

(Here $\cos\alpha > 0$: a rifle pointing straight up never reaches the launcher, since $l > 0$.) A hit needs $y_B = y_C$ at this moment. The $-\frac{1}{2}gt^2$ terms cancel, so

$$(u\sin\alpha)\,t = \left(vT - \tfrac{1}{2}gT^2\right) + (v - gT)\,t.$$

Substituting $t = l/(u\cos\alpha)$ gives

$$\frac{l\sin\alpha}{\cos\alpha} = \left(vT - \tfrac{1}{2}gT^2\right) + \frac{(v - gT)\,l}{u\cos\alpha},$$

and multiplying through by $\cos\alpha$,

$$l\sin\alpha - \left(vT - \tfrac{1}{2}gT^2\right)\cos\alpha = \frac{v - gT}{u}\,l,$$

which is the required result.

The case $T = 0$

With $T = 0$ the condition becomes $l\sin\alpha = vl/u$, and since $l > 0$,

$$\sin\alpha = \frac{v}{u}.$$

? **Ask yourself** Why does this force $v < u$, and not just $v \le u$?

We need $\sin\alpha \le 1$, so $v \le u$. But $v = u$ would need $\alpha = \frac{\pi}{2}$, and then $\cos\alpha = 0$: the bullet goes straight up from O and never reaches the line $x = l$. So a hit is possible only if $v < u$. Then

$$\cos\alpha = \sqrt{1 - \frac{v^2}{u^2}} = \frac{\sqrt{u^2 - v^2}}{u},$$

taking the positive root because $0 < \alpha < \frac{\pi}{2}$, and the hit happens at

$$t = \frac{l}{u\cos\alpha} = \frac{l}{\sqrt{u^2 - v^2}}.$$

The clay is in the air from $t = 0$ until $vt - \frac{1}{2}gt^2 = 0$ again, that is, until $t = 2v/g$. Hitting it before it lands therefore requires

$$\frac{l}{\sqrt{u^2 - v^2}} < \frac{2v}{g} \quad \Longleftrightarrow \quad \frac{2v\sqrt{u^2 - v^2}}{g} > l,$$

where we multiplied by the positive quantity $\sqrt{u^2 - v^2}$. Together with $v < u$, this is what we had to show.

There's a nice way to see the $T = 0$ case. With $\sin\alpha = v/u$ the bullet's vertical velocity component is $u\sin\alpha = v$, the same as the clay's, and both feel the same gravity. So the bullet and the clay are at the *same height at every instant*, and the only question is whether the bullet's horizontal speed $\sqrt{u^2 - v^2}$ covers the distance l before the clay comes down.

Checks

- *Gravity-free limit.* If $g = 0$ and $T = 0$, both objects move in straight lines, and the condition $\sin\alpha = v/u$ is just the usual "aim off" condition for an interception at constant velocity.
- *Maximum range.* For fixed u, the bound $2v\sqrt{u^2 - v^2}/g$ is largest when $v = u/\sqrt{2}$, where it equals u^2/g, the maximum range of a projectile fired at speed u. That makes sense: with $v = u/\sqrt{2}$ we get $\alpha = 45^\circ$, and the hit happens just as the bullet would come back to the ground.
- *Numbers.* With $g = 9.8$, $v = 20$, $T = 0.7$, $u = 120$ and $l = 60$ (SI units), solving for the hitting angle numerically gives $\alpha \approx 0.2987$ rad, and both sides of the first identity agree. Stepping the two motions forward in time, the bullet and clay are both at height 17.13 m when the bullet reaches the launcher. For $T = 0$, $v = 20$ and $u = 100$, the bound is about 400 m with $\alpha \approx 11.5^\circ$.

Common slips

- Using the same time variable for both objects without allowing for the delay T: the clay's height must use $T + t$.
- Forgetting to rule out $v = u$, which gives $\alpha = 90^\circ$ and no hit at all.
- Checking only that the bullet reaches the launcher, and not that the clay is still in the air at that moment.

Beyond STEP

The cancellation of $\frac{1}{2}gt^2$ is a special case of a deep idea: in a frame falling freely with acceleration g, gravity disappears and every projectile moves in a straight line at constant velocity. This is Galilean relativity applied to an accelerating frame, and in

Einstein's hands it became the equivalence principle at the heart of general relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and non-inertial frames carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.